

# Ketan Tiwari

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Portfolio

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## EDUCATION

### Shah & Anchor Kutchhi Engineering College

Bachelor of Technology - Electronics and Computer Science; CGPA: 7.87

Mumbai, India  
2022-2026

## SKILLS SUMMARY

- **Programming Languages** Python, Java, Javascript
- **Databases** PostgreSQL, NoSQL
- **Tools & Platforms** Git, GitHub, Visual Studio Code, Jupyter Notebook, Google Colab
- **AI & Machine Learning** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Generative AI (GenAI), Machine Learning, Deep Learning, Natural Language Processing (NLP)
- **Software Development** Backend Development, Frontend Development, REST APIs, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Software Development Life Cycle (SDLC)
- **Professional Skills** Problem Solving, Analytical Thinking, Communication, Team Collaboration

## EXPERIENCE

### ONGC

Web Developer Intern

Mumbai, India  
May 2025 - July 2025

- Designed and developed dynamic forms with input validation to ensure accurate and structured data collection.
- Implemented structured database storage and automated backend workflows to reduce manual entry errors and improve data accuracy.
- Built search and filtering mechanisms for efficient data querying and retrieval.

### Sapphire Infocom Pvt Ltd

Backend Developer Intern

Mumbai, India  
Sep 2025 - Dec 2025

- Developed scalable backend systems using Node.js and Express.js for structured data processing and management.
- Built RESTful APIs and integrated MongoDB for efficient data exchange, storage, and retrieval.
- Implemented JWT-based authentication to secure API endpoints and protect user data.

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## PROJECTS

### • Natural Language Prompt to SQL Query Generator

- Developed an AI-powered Text-to-SQL chatbot using LLMs to convert natural language into SQL queries for enterprise databases.
- Applied statistical analysis and Exploratory Data Analysis (EDA) on SQL query results to generate insights through interactive charts and visualizations.
- Evaluated SQL query accuracy using F1-Score, Recall, MSE, and RMSE to measure model performance.

### • Multi-Agent AI Research System

- Built an autonomous multi-agent research pipeline using LangChain, Tavily, and BeautifulSoup for web search, content extraction, and automated report generation.
- Implemented LCEL-based Writer and Critic agents for research synthesis, quality evaluation, and iterative refinement.
- Developed a Streamlit UI to initiate research, track agent progress, and export structured research reports.

### • VISTA – AI Platform

- Designing and developing VISTA, an all-in-one AI platform integrating multiple AI capabilities including conversational AI, video intelligence, document analysis, image understanding, code assistance, data analysis, speech-to-text, and AI agents..
- Built intelligent AI workflows using Python, LangChain, RAG, OpenAI, Mistral AI, Sarvam AI, and HuggingFace, enabling automated content understanding, reasoning, summarization, and structured output generation.
- Implemented AI-powered agents and automation workflows that can perform multi-step tasks such as resume analysis, job requirement matching, opportunity search, comparison, and shortlisting through an interactive unified interface.

### • AI Interview Prep & Resume Analytics Platform

- Developed a Full-Stack AI Interview Platform using Node.js, Express, and React/SCSS, featuring a modular, responsive dashboard for mock interview workflows.
- Integrated Generative AI APIs to dynamically analyze user inputs, generate tailored technical questions, and deliver real-time, context-aware performance feedback.
- Engineered a Server-Side Resume Engine that programmatically parses user profiles and generates downloadable, professionally styled PDF documents.